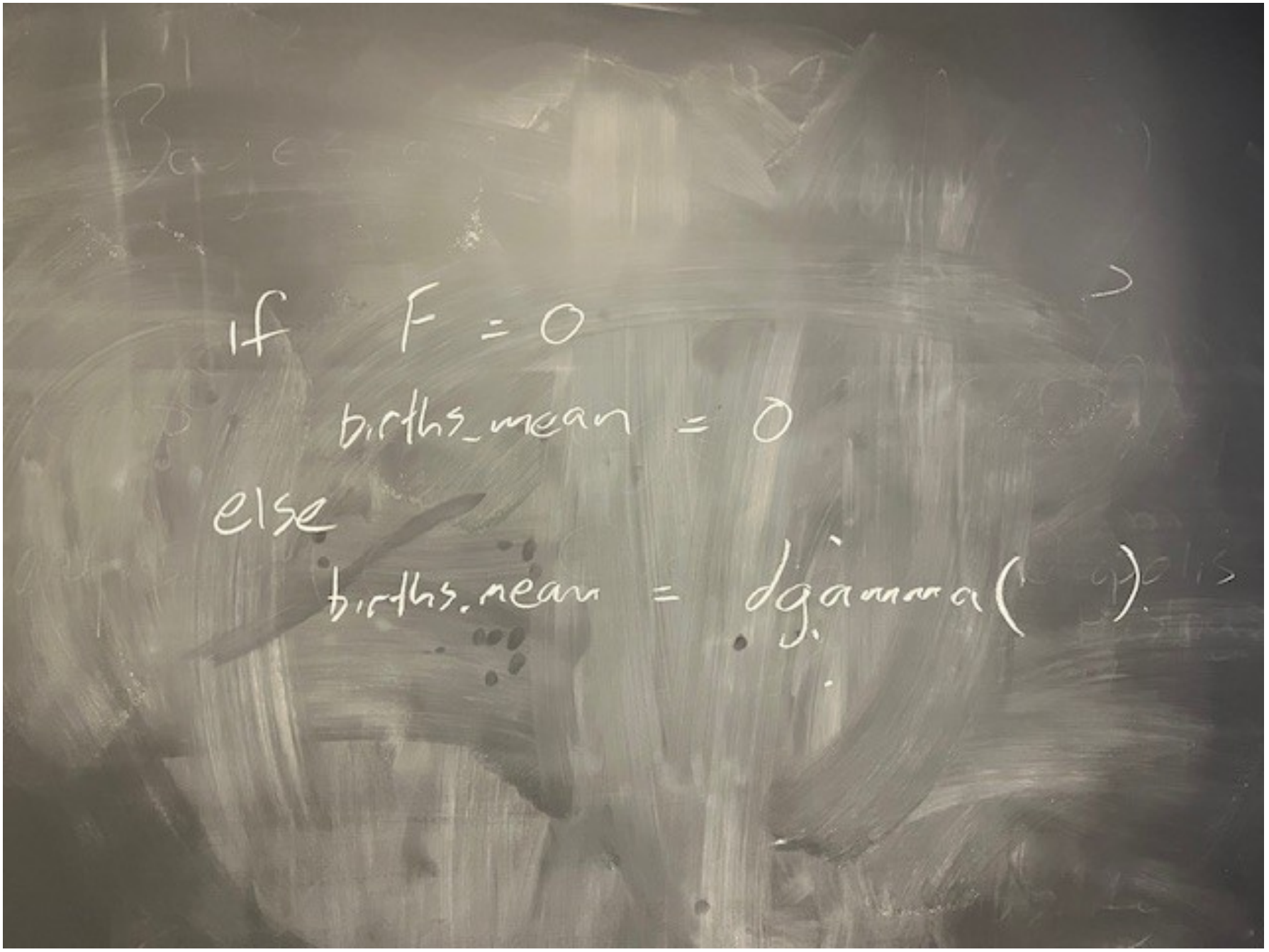

if F = 0
births_mean = 0
else
births_mean = dgamma(

DGP model

Marginalizing

For one population (data point)

$N_t = 2$.

$F \sim \text{Binomial}(N_t, p)$ $p = 0.5$.

0, 1, 2.

$B|F \sim \text{Poisson}\left(\frac{R}{P} F\right)$

DD surv

$(N_{t+1}|B) \sim \text{Binomial}\left(B, e^{\frac{-\alpha N_t}{DD}}\right)$

$\therefore N_{t+1}|F \sim \text{Poisson}\left(\frac{R}{D} F e^{-\alpha N_t}\right)$

$$Pr(N_{t+1}) = \sum_{f=0}^{N_t} P(F=f|N_t) P(N_{t+1}|F, N_t)$$